Srikrishnaa Vadivel

Machine Learning Engineer | Scientific ML | Equivariant Models

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Summary

PhD researcher applying machine learning to first-principles materials and molecular systems. Experience spans PyTorch, equivariant graph networks, diffusion models, distributed training concepts, HPC data pipelines, and physics-based loss design.

ML Experience

Yale University, PhD Researcher 2021–present

- Developed first-principles datasets and workflows for self-trapped exciton calculations on leadership-class HPC resources.
- Prototyped EGNN-style force models with energy gradients $F_{ai} = -\partial E/\partial R_{ai}$ and molecular graph construction.
- Built diffusion-model notes and code scaffolds covering DDPM training, conditional inference, transformers, and index-notation derivations.

Probe Technologies / Weatherford 2019–2021

- Built GPU-accelerated processing and visualization for large acoustic waveform data streams using CUDA and OpenGL.
- Engineered production data tooling around ASP services and Microsoft database systems.

Selected Projects

- **EGNN QM9 Force Model**: PyTorch message-passing model, force loss, synthetic graph fallback, and training-curve generation.
- **Diffusion Notes**: DDPM equations with α_t, β_t, conditional training, and inference in index notation.
- **LiF Band Scaffold**: reciprocal-space Hamiltonian assembly and eigenvalue plotting for physics-informed ML context.

Education

Yale University, PhD Electrical Engineering, 2021–present
Cornell University, MEng ECE, 2017–2018
Cornell University, BA ECE, 2013–2017

Skills

PyTorch, NumPy, SciPy, pandas, scikit-learn, Hugging Face Transformers, Accelerate, datasets, CUDA, MPI, OpenMP, Linux, Python, C++, Docker, Kubernetes, PETSc, SLEPc.